Jonatan David Villalba

Senior Full-Stack Engineer · TypeScript | Node.js · NestJS · PostgreSQL · React · Next.js · Angular
AI product features: LLMs, agents, MCP · Backend-for-Frontend · SAP Composable Storefront (Spartacus)
Open to Remote (US / LATAM) — UTC-3, full ET overlap

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Professional Summary

Senior Full-Stack Engineer with 10 years shipping production web products, now focused on the backend with NestJS. At Qubik Commerce I led the frontend of high-traffic e-commerce storefronts for Crocs, Belsport, Bold and Marchand, built in Angular on SAP Composable Storefront (Spartacus), and worked across the stack: Node.js Backend-for-Frontend services, REST and GraphQL integration, PostgreSQL queries and migrations, Redis caching and server-side auth. I also built the AI layer of SafeCore, Qubik's multi-tenant risk-analysis product (Vercel AI SDK, GPT-4o mini, RAG with pgvector, evals), and an MCP server that gave coding agents the codebase's business rules, cutting cross-team clarification questions by 30%+. Since mid-2026 I have been building NestJS 12 services with Prisma and PostgreSQL — JWT auth with guards, OpenAPI, integration tests with Testcontainers, BullMQ queues and OpenTelemetry — and a Next.js 16 + NestJS e-commerce with Mercado Pago checkout. Earlier, as a freelancer, I owned backends end to end on Node.js, Express and PostgreSQL. Strong in TDD and CI/CD, Design Systems, Core Web Vitals and WCAG 2.1 AA. English: professional working proficiency.

Experience

Qubik Commerce — Buenos Aires, Argentina · Remote
Senior Software Engineer · Full-Stack & AI features — Mar 2021 – Jun 2026

- Led the frontend of high-traffic e-commerce storefronts for brands including Crocs, Belsport, Bold and Marchand, built in Angular (up to v21, Signals, SSR) on SAP Composable Storefront (Spartacus) and SAP Commerce Cloud.
- Worked across the stack inside an established platform architecture: implemented and maintained Node.js Backend-for-Frontend services and Next.js API Route Handlers to the contracts defined by the platform team, aggregating GraphQL and REST endpoints from distributed microservices into typed responses for the UI, with schema validation at each boundary.
- Built the AI layer of SafeCore, Qubik's multi-tenant risk-analysis product, with the Vercel AI SDK and GPT-4o mini: streaming responses, tool calling and Zod-validated structured output server-side in Node.js, retrieval-augmented generation (RAG) with pgvector, and evals to check answer quality before changes shipped.
- Built an internal MCP (Model Context Protocol) server that exposed the repository's business rules, backend and frontend architecture and library conventions to AI coding agents, so answers came from the codebase instead of from people.
- Led the team's adoption of AI-assisted engineering on top of it (Claude Code, Cursor, shared project rules and review standards for AI-generated code), cutting clarification questions between engineering, QA and business by at least 30%.
- Implemented resilience in the BFF — per-call timeouts, bounded retries with backoff and explicit degradation paths — so a slow downstream microservice returned a usable response instead of taking the storefront down.
- Implemented server-side authentication against the platform's identity provider (OAuth2, JWT, refresh rotation, HttpOnly cookies), plus Redis and HTTP caching that cut redundant upstream calls on catalog endpoints; wrote PostgreSQL queries and migrations and tuned indexes when latency regressed.
- Engineered Nx monorepos and published internal component libraries and Design Systems for Angular 21 and React/Next.js 16 (Tailwind CSS, Angular Material, PrimeNG), consumed by multiple squads and accelerating delivery by 40%.
- Drove TDD (Vitest, Jest, Cypress) and CI/CD (GitHub Actions, Jenkins) with quality gates on every deploy; optimized Core Web Vitals (LCP, CLS, INP), improving initial load speed by 35%, and enforced WCAG 2.1 AA.

Onikom LATAM — Remote · Mexico · Contract (concurrent)
Frontend Tech Lead & Solution Architect — Nov 2024 – Apr 2025

- **Led the frontend** architecture and delivery of enterprise storefront solutions on SAP Composable Storefront (Spartacus), Angular 17+ and TypeScript.
- Defined frontend architectural governance, microfrontend patterns and reactive state conventions (NgRx / Signals), and acted as the technical counterpart to product, **backend (Java / Node.js)** and UX/UI — **negotiating API contracts and payload shapes** with the backend team so the frontend was not forced to reshape data it should have received ready to render.
- Resolved rendering bottlenecks, optimized SSR hydration, and ensured decoupled integration between frontend services and enterprise microservices.
- Ran technical plannings, defined interface contracts and conducted code reviews enforcing enterprise quality standards.

Onikom LATAM Remote · Mexico · Contract (concurrent)
Frontend Tech Lead & Solution Architect Feb 2023 – Jul 2023

- Architected the baseline frontend foundation for a high-traffic client storefront, establishing reusable Angular component systems and robust API integration layers.
- Wrote comprehensive Technical Design Documents (TDD), mapped complex functional requirements to maintainable frontend contracts, and oversaw sprint execution.
- Coached engineering squads on reactive programming paradigms with RxJS, modular folder structures, and automated unit testing best practices.

JDV Developer Buenos Aires, Argentina · Remote
Full-Stack Engineer (Freelance) Mar 2016 – Mar 2021

- Delivered custom full-stack web and mobile applications end to end — schema design, REST API, auth, deploy and UI — with React, Next.js, Angular, **Node.js, Express, PostgreSQL**, MySQL and MongoDB.
- Owned the server side of each project: relational data modelling, migrations, authentication and authorization, and deployment, working directly with clients on requirements and trade-offs.
- Built responsive, mobile-first interfaces and progressive web apps with modern JavaScript (ES6+), CSS3/SCSS and Tailwind CSS, consuming secure RESTful endpoints.
- Integrated end-to-end checkout and payment flows with the Mercado Pago Checkout Pro API, ensuring transactional integrity and conversion optimization.

Projects

links-api — production-shaped NestJS API 2026 – in progress
NestJS 12 · TypeScript · Prisma 7 · PostgreSQL · Zod · Vitest + Testcontainers · OpenAPI · BullMQ/Redis · OpenTelemetry · Docker. Built step by step: persistence behind a port, JWT auth with a guard and a @CurrentUser() decorator, integration tests against a real Postgres, OpenAPI generated from the Zod schemas, background jobs with retries, tracing and metrics, and a multi-stage image with graceful shutdown.

E-commerce — Next.js 16 + NestJS 12 2026 – in progress
Next.js 16 (App Router, Server Actions) · NestJS 12 · Prisma · PostgreSQL · Redis · Mercado Pago. Catalog, cart and checkout with the session owned by the API (JWT in an HttpOnly cookie), order snapshots, payment webhooks, full-text search and AI-generated product descriptions with streaming.

Wildfire Risk Agent — TypeScript tool loop
TypeScript · Vercel AI SDK · Zod · tool loop with step cap and per-tool timeouts · Vitest. An agent that computes wildfire-risk indices (Chandler Burning Index / McArthur FFDI) from station weather data, chooses the index the available data can actually support, and throws an explicit, causal error rather than returning a confident wrong number when an input is missing.

Medusa.js Enterprise Storefront
Next.js 15 (App Router) · Medusa.js v2 · React 19 · TanStack Query · Tailwind CSS · TypeScript · BFF Pattern

CineScope
Angular 21 · Clean Architecture (5 layers) · Angular Signals + @ngrx/signals · RxJS · SSR · PWA
cinemascope-jdv.vercel.app

Portfolio — jonatandvillalbaweb.com.ar
Next.js 16 · Tailwind CSS v4 · TypeScript · Zod · next-intl (EN/ES). Migrated from an Angular SSR app: 105 statically prerendered routes, typed content validated with Zod, hreflang and sitemaps, 154 automated tests (unit, integration and Playwright E2E).
jonatandvillalbaweb.com.ar

Skills

Backend, APIs & Data
NestJS (modules, DI, guards, pipes, interceptors, OpenAPI), Node.js, Express, Prisma, PostgreSQL, Redis, BullMQ, REST API design, GraphQL, Backend-for-Frontend (BFF), Next.js Route Handlers, schema design & migrations, authentication (OAuth2, JWT, HttpOnly cookies, sessions), rate limiting, caching, Testcontainers, OpenTelemetry, Docker, MongoDB, Drizzle/ TypeORM

AI Engineering (TypeScript)
LLM integration in TypeScript (Vercel AI SDK, provider SDKs): streaming, tool calling, structured output with Zod; agents with step limits; MCP servers and clients; RAG (pgvector, hybrid retrieval); generative UI; evals and cost/latency budgets; prompt-injection defence. Coding agents: Claude Code, Cursor.

Frontend & Architecture

React, Next.js (App Router, SSR/SSG/ISR, RSC), Angular (v8–v21, Signals, Standalone), Micro Frontends, Monorepos (Nx, Lerna), Design Systems, Tailwind CSS, PWA, i18n

Languages & Core

TypeScript, JavaScript (ES6+), Node.js, SQL, HTML5, CSS3/SCSS

State Management & Reactivity

TanStack Query (React Query), Zustand, Redux Toolkit, Context API, NgRx (Store, Effects, SignalStore), Angular Signals, RxJS

Testing, Quality & DevOps

TDD, Vitest, Jest, Jasmine, Karma, Cypress, React Testing Library, testing pyramid, CI/CD (GitHub Actions, Jenkins), Git/GitFlow, Docker, observability & structured logging, Core Web Vitals, WCAG 2.1 AA accessibility, Scrum/Kanban

E-commerce & Enterprise Platforms

SAP Composable Storefront (Spartacus), SAP Commerce Cloud (CCv2), SmartEdit, Backoffice, Headless/Composable Commerce, Medusa.js v2, Mercado Pago Checkout Pro

Education

Universidad Nacional de José Clemente Paz (UNPAZ) Systems Information Analyst (3-year University Degree) • GPA 8.72/10
Systems Analysis Buenos Aires, Argentina • 2019 – 2023

Certifications

Claude 101 Certification Apr 2026
Anthropic
verify.skilljar.com/c/tbej43odnzb4

Introduction to Agent Skills Mar 2026
Anthropic
verify.skilljar.com/c/nr797hzxr34t

Claude Code in Action Mar 2026
Anthropic
verify.skilljar.com/c/r6xw5e6oo9rs

EF SET English Certificate 65/100 (C1 — reading & listening) Aug 2026
EF Standard English Test (EF SET)
cert.efset.org/es/ScvWpZ

Node.js (Intermediate) Certificate Aug 2026
HackerRank
hackerrank.com/certificates/09ffe2579635

Frontend Developer (React / JavaScript / CSS) Jul 2026
HackerRank
hackerrank.com/certificates/c47600d69728

Server-Side Development with NodeJS, Express and MongoDB Oct 2020
Coursera / Universidad Autónoma de Madrid
coursera.org/account/accomplishments/certificate/VHFUMEB7WBXN

Languages

English
Professional working proficiency — C1 reading & listening (EF SET 65/100)

Spanish
Native

Profiles

LinkedIn
https://www.linkedin.com/in/jonatan-david-villalba